Cairnly Assistant

Customer support chatbot

Operation of Large Language Models

PROJECT 1 / GROUP A
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Problem, scope and knowledge sources

Cairnly Outdoor is a fictional EU outdoor-gear shop. We built a working prototype for recurring support questions.

Customer requests

- Shipping, returns and warranty
- “Where is my order?”
- Questions with exceptions
- Category (7) and urgency classification

Simulated knowledge

- 25 Help Centre documents
- 5 invented orders
- Read-only SQLite database
- Rebuilt by data/create_db.py

Project boundaries

- Answer only from supplied data
- Decline unsupported requests
- Never handle payment cards
- No real personal or order data

Evidence basis: synthetic policies include 30-day returns, warranty rules and non-returnable / restricted goods.

Implemented LLM workflow

The demo and the evaluation use the same chatbot.py code.

01 INPUT SAFETY

Mask card numbers, e-mails and phones; block known attack phrases.

→

02 RAG SEARCH

Multilingual vectors retrieve the 3 closest Help Centre documents.

→

03 ANSWER MODEL

gpt-oss-120b on Groq; fixed grounded prompt; temperature 0.2.

→

04 CHECK OUTPUT

Validate JSON: category, urgency, answer and sources; final safety filter.

Conditional tool: one fixed, read-only SQLite lookup by order number only (no e-mail needed, no writes).

Retrieving 3 vs. 25 documents cut context tokens by 86% (316 vs. 2,329); e5 had the best MRR of 6 retrievers (0.90).

Evaluation design

34

predefined cases in 10 groups

Cases written up front; revised only when the design changed.

PASS RULE

Expected answer or refusal

All required facts present

Nothing forbidden

Correct tool decision

CASE COVERAGE

Policies and exceptions

4 order lookups

Typo + German query

Injection, off-topic, card data

OTHER METRICS

Category / urgency accuracy

6 retrieval strategies

Guardrails-off ablation

gpt-oss-20b judge (1–5)

Results and diagnostics

33 / 34

cases passed (97%)

Good small-scale evidence of a functioning chatbot; not proof of readiness for real customers.

STRUCTURE

100% valid JSON

100% tool decisions

8.1 s mean latency

1,604 prompt + 103 output tokens per answer

QUALITY

Category: 97%

Urgency: 88%

Judge: 5.00 / 5 correct

5.00 / 5 grounded

4.59 / 5 helpful

FAILURE ANALYSIS

T30: keyword grader wrongly rejected a safe card refusal.

Guardrails on/off: 5/6 vs 4/6 attack + off-topic cases passed.

Order lookups: 16–24 s.

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Limitations and next steps

97% on synthetic tests is not deployment readiness.

TEST COVERAGE	34 self-written cases; limited safety and multi-turn coverage.
PRIVACY	Order number alone reveals status; CVV and expiry are not masked.
ROBUSTNESS	Fixed list of 9 attack phrases; nested tags bypass tag stripping.
RETRIEVAL & TOOL	No similarity threshold: top 3 always sent; no tool-call limit.

PROJECT 2 / GROUP A

Red-team and improve safety.

Priorities: verify order ownership, strengthen redaction, probe unseen injections and add escalation.

Course prototype only • simulated data • no live rollout